Retinex-Based Dual-Branch Feature Extraction Network for Hyperspectral Image Classification

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***Abstract*—Deep learning-based methods for hyperspectral image classification (HSIC) have been widely utilized in recent years. However, existing HSIC methods do not adequately account for illumination variations in HSIs, particularly in urban areas where shadows created by complex ground objects result in significant variations that cannot be ignored. Additionally, when dealing with limited labeled data, most deep learning methods are prone to overfitting, resulting in poor classification performance. To address these challenges, we propose a new Retinex-based dual-branch feature extraction network (RDFEN) for HSIC. First, by incorporating Retinex theory, we propose a hyperspectral Retinex (HyperRetinex) module to extract illumination attributes and reflectance attributes. Then, we propose a dual-branch feature extraction (DBFE) network, which consists of two submodules: illumination attributes feature extraction (IAFE) module and reflectance attributes feature extraction (RAFE) module. Finally, an illumination–reflectance attributes interaction attention fusion (IRAIAF) module is strategically designed to integrate distinct features. Experiments on four benchmark HSI datasets demonstrate that the proposed method outperforms other state-of-the-art HSIC methods, achieving up to 96.85% overall accuracy (OA) on the Pavia University (PU) dataset, thereby highlighting its effectiveness and robustness in HSIC. For reproducibility, the code is available at https://github.com/JT-shen/Code**

***Index Terms*—Dual-branch feature extraction (DBFE) network, hyperspectral image classification (HSIC), interactive attention mechanism, Retinex model.**

I. Introduction

HYPERSPECTRAL imaging has emerged as a transformative advancement in the remote sensing community. Hyperspectral images (HSIs) commonly include hundreds of narrow spectral bands, with a spectral resolution of a few nanometers, enabling the precise characterization of material-specific spectral signatures for ground objects. Benefiting from their unique attributes, HSIs find extensive applications across a multitude of domains, including medical diagnostics [1], [2], military reconnaissance [3], [4], ocean observations [5], [6], and so on. In virtually all of these domains, HSI classification (HSIC) is a critical task, focusing on pixel-wise classification to assign semantic labels that enable fine-grained object recognition.

During the initial stages of HSIC research, methods such as the support vector machine (SVM) [7], [8], random forest (RF) [9], and sparse representation (SR) [10] were used. These methods are constrained by their reliance on prior knowledge, limited feature representation capabilities, and suboptimal classification performance. To effectively exploit the spatial information contained in HSIs, numerous spectral–spatial methods have been developed, including superpixel-based methods [11], [12], Gabor filter-based methods [13], edge-preserving filtering-based methods [14], [15], [16], and so on. However, the representational capacity of these methods is limited, as they are critically dependent on hand-crafted parameter configurations requiring manual tuning.

In recent years, deep learning-based methods have driven remarkable advancements in HSIC [17], [18], [19]. Pioneering deep learning methods, such as deep autoencoders (DAEs) [20] and deep belief networks (DBNs) [21], were employed to perform dimensionality reduction and feature extraction from HSIs via hierarchical nonlinear transformations, establishing foundational frameworks for modern architectures. Thereafter, convolutional neural networks (CNNs) [22], [23] have been widely employed for HSIC, primarily due to their success in image recognition tasks [24]. CNNs can effectively capture spatial and spectral features through multilevel convolution and pooling operations, thus improving classification accuracy and robustness. To harness integrated spectral–spatial features in HSIs, CNN architectures with three-channel inputs were adapted for HSIC [25]. In [26], a dual-branch CNN architecture was proposed to extract joint spectral–spatial features from HSIs. The integration of CNNs with other technologies for HSIC has also been widely investigated. In [27], SimAM-based CNNs were integrated with Transformers to achieve satisfying classification accuracy and computational speed simultaneously. In [28], a CNN–Transformer fusion network was proposed, which leverages the strengths of CNNs in extracting local features and the Transformer's capability

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for long-range dependency modeling in HSIC. Furthermore, graph convolutional networks (GCNs) have enhanced the comprehension of intricate relationships in HSIs by effectively capturing and modeling complex graph-based patterns [29]. In [30], a global–local GCN-based feature fusion network with rapid training convergence was introduced. Recurrent neural networks (RNNs) [31] and long short-term memory networks (LSTMs) [32] have proved to be effective in modeling long sequences of spectral data, further enriching the toolkit for HSIC. More recently, Transformers and Mamba have emerged as dominant tools in the task of HSIC [33]. In [34], a tensor-based Transformer network was proposed, preserving the intrinsic structural dependencies in the samples. In [35], a hybrid CNN–Transformer architecture was developed for characterizing the surface of Mars, jointly modeling local–global features present in HSIs. In [36], an interval group spatial–spectral Mamba-based model was proposed, enabling multidirectional and multiscale information extraction. To alleviate the computational complexity, a lightweight dual-branch Mamba-convolution network was proposed [37]. While existing methods claim to effectively model spectral–spatial features in HSIs, several domain-specific limitations still persist.

1) Existing HSIC methods do not adequately account for illumination variations in HSIs. Illumination variation is a critical bottleneck for HSIC, particularly in urban environments where dense building structures induce complex shadow geometries that cause nonlinear spectral distortions. These distortions are manifested as nonuniform attenuation across spectral bands and disrupted absorption features, severely compromising material discriminability. Uneven illumination further exacerbates the Hughes phenomenon by introducing illumination-specific variance that dominates the spectral feature space, thereby reducing model generalization capability.
2) DL-based methods heavily depend on extensively labeled datasets for training. However, the process of annotating HSIs is labor-intensive and requires expert involvement due to their complex spectral characteristics.

To address these challenges, we propose a new Retinex-based dual-branch feature extraction network (RDFEN) for HSIC. First, by incorporating Retinex theory, we propose a hyperspectral Retinex [HyperRetinex (HR)] module to extract illumination attributes and reflectance attributes. Then, we propose a dual-branch feature extraction (DBFE) network, which consists of two submodules: illumination attributes feature extraction (IAFE) module and reflectance attributes feature extraction (RAFE) module. Finally, an illumination–reflectance attributes interaction attention fusion (IRAIAF) module is strategically designed to integrate distinct features.

Overall, this article makes the following innovative contributions.

1) We propose an HR module that can extract illumination attributes and reflectance attributes from HSIs. To the best of our knowledge, this is the first work successfully adapting Retinex theory to HSIC problems.
2) We design a DBFE network, consisting of an IAFE module and an RAFE module. The IAFE module isolates illumination variations (e.g., shadows), while the RAFE module extracts intrinsic reflectance properties of materials. This dual-branch structure decouples illumination-induced distortions from material-specific features, enhancing discriminative power and mitigating spectral–spatial misclassifications under varying illumination conditions.
3) We develop an IRAIAF module, which effectively enhances intercomponent interactions in HSIs. The IRAIAF enables the model to capture intrinsic relationships between different components, significantly improving classification performance in HSIC problems.

Experiments on four datasets demonstrate that the proposed method robustly disentangles intrinsic and environmental features in HSIs, significantly improving classification performance. Illumination variations are particularly pronounced in urban scenes with human-made structures, where complex illumination conditions induce spectral distortions, degrading classification accuracy. The proposed method exhibits strong robustness to illumination variations, yielding notable performance gains in urban environments under complex illumination scenarios.

The remainder of this article is organized as follows. Section II provides the background of our research. Section III describes the proposed method in detail. In Section IV, a comprehensive evaluation is conducted through experiments on four datasets to validate the effectiveness of the proposed method. Finally, Section V presents key findings and their implications for future research.

II. Related Works

In this section, we provide a brief overview of two methodologies utilized for constructing our model.

A. Retinex Method

The Retinex model is a relevant concept in computer vision, primarily focusing on image enhancement and color constancy. The term “Retinex” is derived from “retina” and “cortex,” illustrating the interplay between retinal photoreceptor responses and cortical signal processing, which collectively influence human color perception. The Retinex model is rooted in the theory that an object’s color perception is determined by its reflectance properties rather than the absolute intensity of reflected light. Reflectance refers to how well an object can reflect light, which varies depending on the object’s material and surface characteristics. In contrast to traditional methods that may focus solely on enhancing certain features of an image (either linearly or nonlinearly), Retinex achieves a balance in dynamic range compression, edge enhancement, and color constancy, making it suitable for adaptive enhancement across various types of images.

The evolution of Retinex algorithms has progressed through several key stages. Land [38] introduced the foundational

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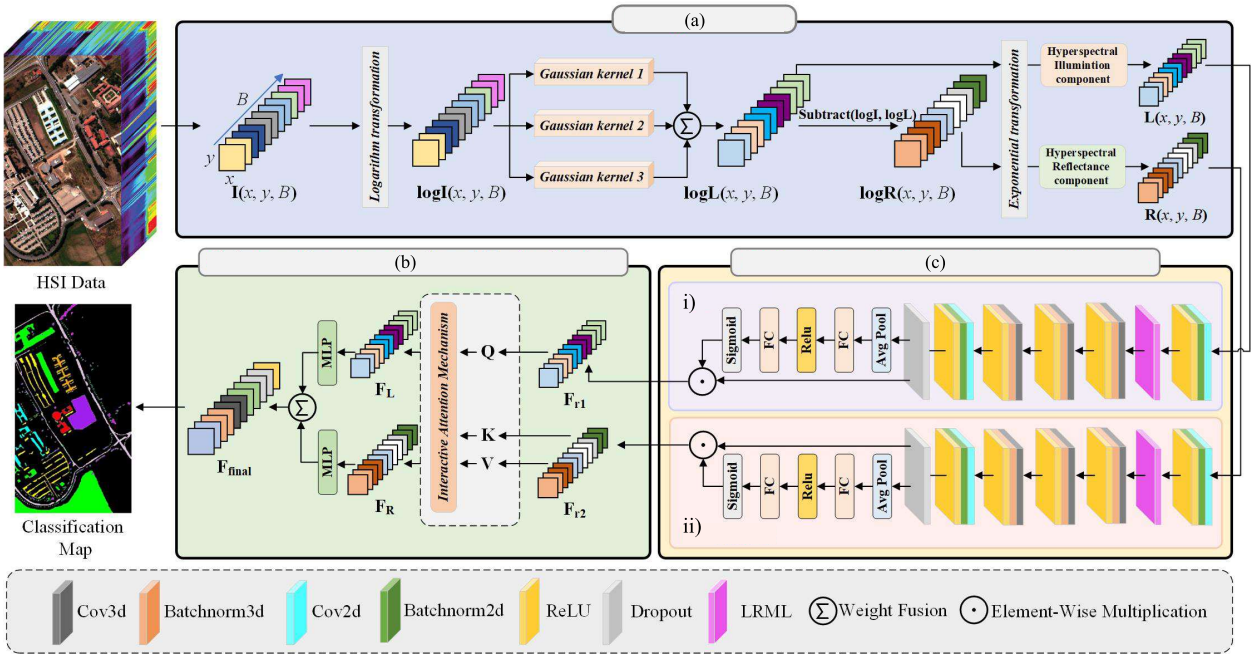


Fig. 1. Proposed RDFEN framework for HSIC. First, the input HSI is decomposed into illumination attributes and reflection attributes via the HR module. Then, the DBFE network processes these representations separately: the IAFE branch captures lighting-invariant features, while the RAFE branch encodes material-specific features. Finally, these complementary features are fused by an interactive attention-based fusion module to enhance discriminative representation, producing the classification result. (a) HR. (b) Dual-branch feature extraction network: i) IAFE and ii) RAFE. (c) Interactive attention-based fusion module.

Retinex theory, which was later developed into the single-scale Retinex (SSR) in the 1980s. The SSR uses a Gaussian filter to smooth images and control enhancement through adjustable filter sizes, enhancing image details effectively. In the 1990s, multiscale weighted average Retinex (MSWAR) [39] was introduced. This method employs filters at multiple scales and combines their results with weighted coefficients for a more natural enhancement. Building on previous multiscale Retinex advancements, the 2000s introduced the color restoration multiscale Retinex [40], which preserved original color information while enhancing low-light images. The evolution of the Retinex algorithm demonstrates a continuous effort to refine image processing under complex lighting conditions, focusing on both enhancement quality and color fidelity. The Retinex theory has been applied to other HSI processing tasks, primarily for image quality improvement. For example, Qu et al. [41] utilized the multiscale Retinex algorithm to extract spatial structure details for fusing hyperspectral and panchromatic images. In contrast, our work is the first attempt to adapt Retinex decomposition specifically for HSIC, transforming it from a tool for image enhancement to a classification-driven feature decoupling strategy.

B. Attention Mechanism

The visual attention mechanism is a key process in the human brain for handling visual signals. Deep learning attention mechanisms draw inspiration from this human process [42]. This mechanism enables humans to selectively focus on relevant details in complex environments, thereby enhancing the efficiency of visual processing with limited resources.

The evolution of HSIC has seen significant advancements through innovative attention mechanisms. Each study builds upon previous work to enhance performance and efficiency. Sun et al. [43] introduced the spectral–spatial attention network (SSAN), which employs attention modules to focus on key spectral and spatial features in HSIs. This model helps to suppress irrelevant pixels, allowing for more efficient processing of important features. Zhu et al. [44] enhanced the spectral attention module and the spatial attention module by embedding them into residual blocks. This modification not only improves model robustness against overfitting but also enhances computational speed, making the model more efficient without sacrificing precision. Wang et al. [45] introduced the multiscale spatial–spectral cross-modal attention network (MS2CANet), which combines multiscale pyramid fusion and cross-modal attention to leverage complementary information from both HSI and LiDAR data. This allows the model to extract richer features, improving classification performance.Existing methods often focus on enhancing features within specific components and fusing cross-modal information. These methods frequently neglect intercomponent interactions within the same modality, which limits the generalization ability of the model. To address this challenge, we developed a novel IRAIAF module, which effectively enhances intercomponent interactions in HSIs. The IRAIAF enables the model to capture intrinsic relationships between different components, significantly improving classification performance in HSIC.

III. Proposed Method

The framework proposed in this article is graphically illustrated in Fig. 1. Our model contains three main modules: the HR module, the DBFE module, and the IRAIAF module. The DBFE module has two submodules: the IAFE module and the RAFE module.

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A. HR Module

Retinex theory aims to elucidate how the human visual system perceives color and brightness consistently, even under widely varying lighting conditions. According to this theory, the human visual system relies not only on the intensity of local pixels but also on the contextual information surrounding them when perceiving color and brightness. Mathematically, the Retinex model can be expressed as follows:

$$\mathbf{I}(x,y) = \mathbf{R}(x,y) \cdot \mathbf{L}(x,y) \tag{1}$$

where $\mathbf{I}(x,y)$ is the pixel value of the input image $\mathbf{I}$ at position (x,y). $\mathbf{R}(x,y)$ is the reflectance image, which represents the intrinsic properties of the object surface. $\mathbf{L}(x,y)$ is the illumination image, which represents the variation of the ambient illumination. The goal is to estimate the reflectance image $\mathbf{R}(x,y)$ from the input image $\mathbf{I}(x,y)$, since the reflectance image is more reflective of the intrinsic properties of the object, while the illumination image contains the effect of the ambient illumination.

In this article, Retinex theory is incorporated into the HSIC task by considering the information of high-dimensional spectral bands [39]. Thus, we propose an HR module, named HR. Let $\mathbf{I}(x,y,B)$ be the HSI and B be the number of spectral bands. Each pixel value of the HSI can be decomposed as the product of reflectance attribute $\mathbf{R}(x,y,B)$ and illumination attribute $\mathbf{L}(x,y,B)$ as follows:

$$\mathbf{I}(x,y,B) = \mathbf{R}(x,y,B) \cdot \mathbf{L}(x,y,B). \tag{2}$$

A Gaussian kernel $G_{\sigma_s}(x,y)$ of the sth scale is used to filter the log-transformed image of each band, as follows:

$$\log \mathbf{L}_{\sigma_s}(x,y,B) = G_{\sigma_s}(x,y) * \log \mathbf{R}(x,y,B). \tag{3}$$

Reflectance attributes at different scales are obtained by subtracting the log-transformed image from the Gaussian filtering results at each scale

$$\log \mathbf{R}_{\sigma_s}(x,y,B) = \log \mathbf{I}(x,y,B) - \log \mathbf{L}_{\sigma_s}(x,y,B). \tag{4}$$

Then, the reflectance attributes for different scales are weighted and averaged to obtain the final reflectance attributes

$$\log \mathbf{R}(x,y,B) = \frac{1}{N_s} \sum_{s=1}^{N_s} \log \mathbf{R}_{\sigma_s}(x,y,B) \tag{5}$$

where N_s denotes the number of scales and is set to 3 in the proposed method. Finally, the final HSI is exponentially transformed to restore the linear space. Finally, the final HSI is exponentially transformed to restore the linear space. The chosen scales ([5, 60, 100] to [35, 120, 400]) were determined through extensive empirical experiments aimed at optimizing classification performance across multiple datasets. These parameters were selected as the best tradeoff between capturing contextual information (larger scales) and preserving fine spectral–spatial details (smaller scales)

$$\begin{aligned} \mathbf{R}(x,y,B) &= \exp\left(\log \mathbf{R}_{\sigma_i}(x,y,B)\right) \\ \mathbf{L}(x,y,B) &= \exp\left(\log \mathbf{L}_{\sigma_i}(x,y,B)\right). \end{aligned} \tag{6}$$

Inspired by feature visualization techniques in scattering networks [46], [47], Fig. 2 demonstrates HR's decomposition capability. The illumination map [see Fig. 2(b)] explicitly captures shadow patterns, while the reflectance component [see Fig. 2(c)] preserves consistent spectral features across illumination conditions. This visual evidence validates our formulation in (2)–(6).



(a) (b) (c)

Fig. 2. Visual results of the HR module on the PU dataset. (a) Input HSI (false-color composite). (b) Estimated illumination component (both a key intermediate step in the Retinex process and the output illumination attribute used for subsequent analysis). (c) Estimated reflectance (final output retaining material-specific spectral properties after illumination separation).

B. DBFE Network

1) IAFE Module: The IAFE module is designed to extract illumination-invariant features from HSIs. Through this module, we aim to mitigate the impact of illumination variations on classification performance.

A six-layer convolutional method is implemented for enhancing local feature extraction. Fig. 1 illustrates that each convolutional layer is accompanied by the related operations. The illumination attributes map $\mathbf{L} \in \mathbb{R}^{H \times W \times B}$ is input into the first layer, where H and W represent the spatial dimensions of the HSI, and B represents the number of spectral bands. First, the HSI patches are fed into 1×1 convolutional blocks to extract spectral information

$$\mathbf{F}^1_{\text{patch}} = \delta\,\text{BN}\left(\text{Conv2d}\left(\mathbf{L}\right)\right) \tag{7}$$

where $\mathbf{F}^1_{\text{patch}}$ is the output of the 1st layer convolution, δ is the activation function [rectified linear unit (ReLU)], and BN is the batch normalization (BN) operation.

To extract illumination-invariant features of objects, we design a lightweight residual modulation layer (LRML). Specifically, a 1×1 convolution is used to extract the intermediate features via a linear transformation, which serves as an abstract representation of the input features. The result of the linear transformation is then mapped to the range $(0,1)$ using a Sigmoid activation function, generating the modulation factor m. Afterward, the original feature map is weighted and reconstructed using this modulation factor, resulting in a new feature representation

$$\mathbf{F}_{\text{i}} = \text{Conv}_{1\times1}\left(\mathbf{F}^1_{\text{patch}}\right) \tag{8}$$

$$m = \text{LF}\left(\mathbf{F}_{\text{i}}\right) = \frac{1}{1+e^{-\mathbf{F}_i}} \tag{9}$$

$$\mathbf{F}^2_{\text{patch}} = \mathbf{F}^1_{\text{patch}} \odot m + \mathbf{F}^1_{\text{patch}} \tag{10}$$

where $\mathbf{F}_{\text{i}}$ represent intermediate features, LF denotes the linear transformation operation, $\mathbf{F}^2_{\text{patch}}$ is the output of the

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TABLE I
DETAILED NETWORK STRUCTURE OF CNN BLOCKS

Layer Number	Type	Size	Normalization	Activation
Layer 1	2D Conv	(1×1)	2D BN	Relu
Layer 2	LRML	(1×1)	-	Sigmoid
Layer 3	3D Conv	(7×3×3)	3D BN	Relu
Layer 4	3D Conv	(5×3×3)	3D BN	Relu
Layer 5	3D Conv	(3×3×3)	3D BN	Relu
Layer 6	2D Conv	(3×3)	2D BN	Relu

second layer convolution, and $\odot$ denotes the element-wise multiplication.

Then, 3-D convolutions are used three times to maintain the spectral information contained in the HSIs. Next, the features extracted from different layers are integrated to extract spatial information using a 2-D convolution layer as follows:

$$\mathbf{F}_{\text{patch}}^{n} = \delta\,\text{BN}\left(\text{Conv3d}\left(\mathbf{F}_{\text{patch}}\right)\right), \quad (n = 3, 4, 5) \tag{11}$$

$$\mathbf{F}_{\text{concat}} = \text{Concat}\left(\mathbf{F}_{\text{patch}}^{2}, \mathbf{F}_{\text{patch}}^{3}, \mathbf{F}_{\text{patch}}^{4}, \mathbf{F}_{\text{patch}}^{5}\right) \tag{12}$$

$$\mathbf{F}_{\text{out1}} = \text{Dropout}\left(\delta\,\text{BN}\left(\text{Conv2d}\left(\mathbf{F}_{\text{concat}}\right)\right)\right) \tag{13}$$

where $\mathbf{F}_{\text{patch}}^{n}$ is the output of the nth layer convolution, $\mathbf{F}_{\text{out1}}$ is the output of the total CNN blocks, and Dropout denotes the dropout operation. The detailed network structure and parameters for the CNN feature extraction modules are provided in Table I.

Finally, the feature maps $\mathbf{F}_{\text{out1}}$ are fed to the channel attention module. The channel attention mechanism uses squeeze and excitation operations to establish relationships between channels of the feature maps. The channel attention weights ω are element-wise multiplied with the feature map $\mathbf{F}_{\text{out1}}$ to obtain the enhanced feature map $\mathbf{F}_{\text{r1}}$.

$$\mathbf{F}_{\text{r1}} = \omega \odot \mathbf{F}_{\text{out1}} \tag{14}$$

During training, the feature extractor is trained using the cross-entropy loss function [48] to minimize the difference between the predicted values and the true labels. The cross-entropy loss function L is defined as follows:

$$L = -\sum_{c=1}^{C} Y_c \log\left(p_c\right) \tag{15}$$

where p is the predicted value, Y is the true label, and C is the number of classes.

2) RAFE Module: The role of the RAFE module is to effectively capture and represent features related to the intrinsic properties of HSIs, enabling accurate classification of different land-cover types. By extracting key features (e.g., material characteristics and texture information) from HSIs, RAFE can enhance the model's representational capabilities. RAFE adopts an identical deep learning architecture to LAFE for feature extraction, and both modules are trained using cross-entropy loss. The mathematical expression is given by

$$\mathbf{F}_{\text{patch}}^{1'} = \delta\,\text{BN}\left(\text{Conv2d}\left(\mathbf{R}\right)\right) \tag{16}$$

$$\mathbf{F}_{\text{patch}}^{2'} = \mathbf{F}_{\text{patch}}^{1'} \odot \text{LF}\left(\text{Conv}_{1\times1}\left(\mathbf{F}_{\text{patch}}^{1'}\right)\right) + \mathbf{F}_{\text{patch}}^{1'} \tag{17}$$

$$\mathbf{F}_{\text{out2}} = \text{Dropout}\left(\delta\,\text{BN}\left(\text{Conv2d}\left(\text{Concat}\left(\mathbf{F}_{\text{patch}}^{2'}, \delta\,\text{BN}\left(\text{Conv3d}\left(\mathbf{F}_{\text{patch}}^{2'}\right)^{\oplus 3}\right)\right)\right)\right)\right) \tag{18}$$

$$\mathbf{F}_{\text{r2}} = w \odot \mathbf{F}_{\text{out2}} \tag{19}$$

where $\mathbf{F}_{\text{patch}}^{1'}$ and $\mathbf{F}_{\text{patch}}^{2'}$ are the outputs of the first and second layer convolutions, respectively. $\mathbf{F}_{\text{out2}}$ is the output of the total CNN blocks and $\mathbf{F}_{\text{r2}}$ is the channel attention enhanced feature.

C. IRAIAF Module

To achieve an effective integration of illumination and reflectance information, the IRAIAF module is proposed to leverage cross-branch dependencies. As shown in Fig. 3, the input feature maps $\mathbf{F}_{\text{r1}}$ and $\mathbf{F}_{\text{r2}}$ are first converted into query, key, and value tensors

$$\begin{aligned}\mathbf{Q} &= \text{Conv 2d}\left(\mathbf{F}_{\text{r1}}; W_q\right)\\ \mathbf{K} &= \text{Conv 2d}\left(\mathbf{F}_{\text{r2}}; W_k\right)\\ \mathbf{V} &= \text{Conv 2d}\left(\mathbf{F}_{\text{r2}}; W_v\right)\end{aligned} \tag{20}$$

where W_q is the convolutional kernel weight for query projection, W_k is the convolutional kernel weight for key projection, and W_v is the convolutional kernel weight for value projection.

Then, an energy matrix $\mathbf{E}$ is computed as the dot product between the query tensor $\mathbf{Q}$ and the key tensor $\mathbf{K}$

$$\mathbf{E} = \mathbf{Q} \cdot \mathbf{K}^{\text{T}}. \tag{21}$$

The Softmax function is applied to generate the attention weight matrix $\mathbf{A}$

$$\mathbf{A} = \mathbf{A}_{ij} = \frac{\exp\left(\mathbf{E}_{ij}\right)}{\sum_{k}^{N} \exp\left(\mathbf{E}_{ik}\right)} \tag{22}$$

where $\mathbf{A}_{ij}$ represents the attention weight from the feature at the ith position to the feature at the jth position and N denotes the number of entries of $\mathbf{E}$.

By applying the attention weight matrix $\mathbf{A}$ to the value matrix $\mathbf{V}$, the resulting feature map $\mathbf{O}$ is obtained, which integrates the complementary information of $\mathbf{F}_{\text{r1}}$ and $\mathbf{F}_{\text{r2}}$

$$\mathbf{O} = \mathbf{V} \cdot \mathbf{A}^{\text{T}}. \tag{23}$$

To maintain feature stability, a learnable scaling parameter γ is introduced, and residual connections are used to combine the new feature map with the original feature maps

$$\begin{aligned}\mathbf{F}_{\text{L}} &= \gamma \cdot \mathbf{O} + \mathbf{F}_{\text{r1}}\\ \mathbf{F}_{\text{R}} &= \gamma \cdot \mathbf{O} + \mathbf{F}_{\text{r2}}.\end{aligned} \tag{24}$$

The enhanced feature maps $\mathbf{F}_{\text{L}}$ and $\mathbf{F}_{\text{R}}$ are processed through two parallel multilayer perceptrons (MLPs) [49], dedicated to decouple spectral discriminability and spatial invariance through illumination and reflectance attributes, respectively,

$$\begin{aligned}\mathbf{L}^{0} &= \mathbf{F}_{\text{L}}, \mathbf{L}^{k} = \sigma\left(\mathbf{W}_{\text{L}}^{k}\mathbf{L}^{k-1} + b_{\text{L}}^{k}\right), \quad k = 1, 2, \ldots, K\\ \mathbf{R}^{0} &= \mathbf{F}_{\text{R}}, \mathbf{R}^{k} = \sigma\left(\mathbf{W}_{\text{R}}^{k}\mathbf{R}^{k-1} + b_{\text{R}}^{k}\right)\end{aligned} \tag{25}$$

where $\mathbf{W}_{\text{L}}^{k}$ and $\mathbf{W}_{\text{R}}^{k}$ represent the weight matrix of the illumination attributes branch and the reflectance attributes branch at the kth layer, respectively, b_{L}^{k} and b_{R}^{k} represents the bias vector of the two branches at the kth layer, σ represents the activation function, K represents the number of layers of the MLP, $\mathbf{L}^{0}$

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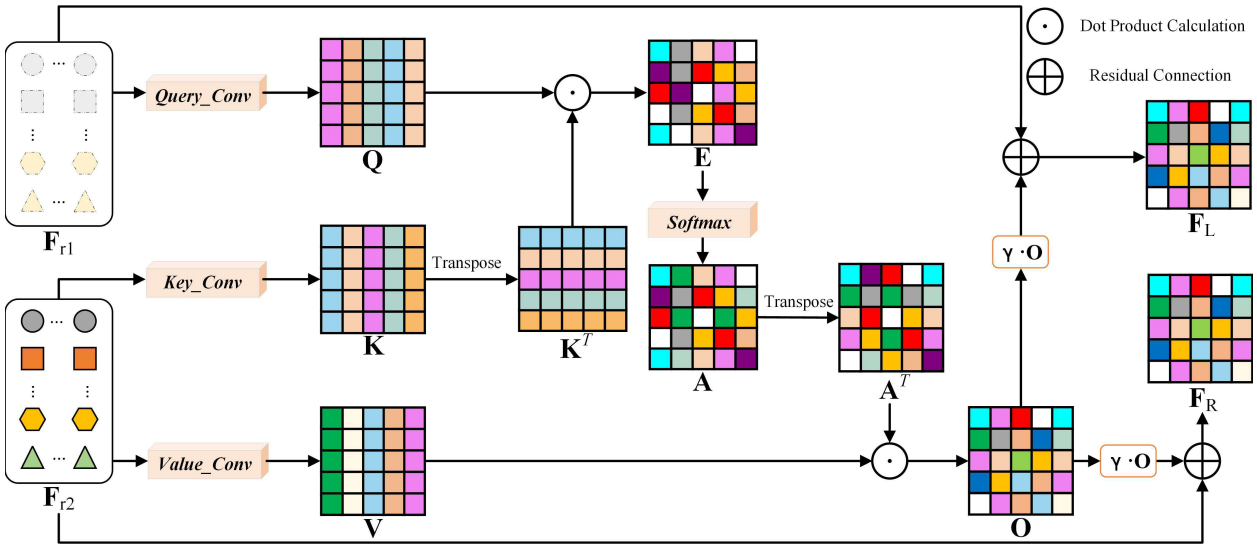


Fig. 3. Structure of the IRAIAF module. First, the input feature maps $\mathbf{F}_{r1}$ and $\mathbf{F}_{r2}$ are transformed into **Q**, **K**, and **V** tensors. Then, the dot product between **Q** and **K** is calculated to obtain the energy matrix **E**, and the Softmax function is applied to generate the attention weight matrix **A**. Then, the value tensor is weighted and summed using the attention weight matrix **A** to obtain the integrated feature map **O**. Finally, the input feature maps are concatenated with the **O** residuals to obtain the output $\mathbf{F}_L$ and $\mathbf{F}_R$.

and $\mathbf{R}^0$ represent the input feature maps of the two branches, respectively, and $\mathbf{L}^k$ and $\mathbf{R}^k$ represent the output feature maps of the two branches after the kth layer, respectively.

In addition, a strategy is designed to integrate the feature representations output by the two MLPs. This strategy uses a weighted summation method at the feature level, and the weights are dynamically learned by the network, which reflects the importance of the features of each branch. This fusion strategy enhances the model's adaptability and feature expression capability.

Weighted fusion can be achieved by the learned weights α_L and α_R, which are used to control the importance of illumination and reflectance features. Therefore, the final fused feature $\mathbf{F}_{\text{final}}$ can be expressed as follows:

$$\begin{aligned} \mathbf{F}_{\text{final}} &= \alpha_L \mathbf{L}^K + \alpha_R \mathbf{R}^K \\ [\alpha_L, \alpha_R] &= \text{softmax}\left([\theta_L, \theta_R]\right) \end{aligned} \tag{26}$$

where θ_L and θ_R are the parameters obtained by the model through learning. The fused feature vectors are fed into a feature discrimination module, whose role is to distinguish samples from different classes. This module is implemented through a series of fully connected layers and finally connected to a classifier that predicts class labels based on the learned feature pattern output

$$P_i = \text{softmax}\left(W_c F + b_c\right) \tag{27}$$

where P_i represents the probability distribution of the categories and W_c and b_c represent the fully connected layer and its bias, respectively.

The IRAIAF module can enhance HSIC performance in two aspects. First, by comparing illumination and reflectance features using the query–key–value framework, the model adaptively assigns higher weights to the most discriminative attributes. For example, in shadowed regions, the reflectance features gain priority to mitigate illumination distortions, while uniform illumination conditions enable balanced fusion. Second, the cross-branch interaction suppresses noise through dual verification. When illumination features are distorted (e.g., by strong reflections), the stable material information from reflectance features dynamically corrects errors. Conversely, in low-contrast areas, spatial context from illumination features aids in boundary localization, enhancing reflectance separability. This complementarity mitigates misclassification at material transition boundaries and spectrally ambiguous regions.

IV. Experimental Analysis

To validate the classification performance of our method, comprehensive experimental analyses on several real datasets are provided in this section.

A. Datasets Description

Comprehensive details about the experimental HSI datasets, that is, Pavia University (PU), Indian Pines (IP), WHU-Hi-LongKou (LK), and Kennedy Space Center (KSC), are provided in Table II. Also, other important information, such as the name, color, and number of classes, is presented in Table III.

B. Experimental Setup

1) Evaluation Metrics: To evaluate the classification performance of our method, several popular metrics, including overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and class accuracy (CA), were adopted.

2) Comparative Methods: Eight advanced HSIC methods were selected for comparison. These methods include the SVM [7] from traditional machine learning, deep learning models

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TABLE II
COMPREHENSIVE DETAILS ABOUT THE EXPERIMENTAL DATASETS

	PU	IP	LK	KSC
Imaging site	Pavia, Northern Italy	North-western Indiana	Longkou Town, China	Kennedy Space Center, USA
Spatial size	610×340	145×145	550×400	512×614
Number of spectral bands	103	200	270	176
Sensor	ROSIS-03	AVIRIS	DJI M600 Pro	AVIRIS
Number of classes	9	16	9	13
Number of samples	42776	10249	204542	5211

TABLE III
NAMES, COLORS, AND NUMBERS OF CLASSES FOR THE EXPERIMENTAL DATASETS

Class No.	PU			IP			LK			KSC		
	Class name	Color	Numbers	Class name	Color	Numbers	Class name	Color	Numbers	Class name	Color	Numbers
1	Asphalt		6631	Alfalfa		46	Corn		34511	Scrub		761
2	Meadows		18649	Corn-notill		1428	Cotton		8374	Willow swamp		243
3	Gravel		2099	Corn-mintill		830	Sesame		3031	Cabbage palm hammock		256
4	Trees		3064	Corn		237	Broad-leaf soybean		63212	Cabbage palm/oak hammock		252
5	Painted metal sheets		1345	Grass-pasture		483	Narrow-leaf soybean		4151	Slash pine		161
6	Bare Soil		5029	Grass-trees		730	Rice		11854	Oak/broadleaf hammock		229
7	Bitumen		1330	Grass-pasture-mowed		28	Water		67056	Hardwood swamp		105
8	Self-blocking bricks		3682	Hay-windrowed		478	Roads and houses		7124	Graminoid marsh		431
9	Shadows		947	Oats		20	Mixed weed		5229	Spartina marsh		520
10	-	-	-	Soybean-notill		972	-	-	-	Cattail marsh		404
11	-	-	-	Soybean-mintill		2455	-	-	-	Salt marsh		419
12	-	-	-	Soybean-clean		593	-	-	-	Mud flats		503
13	-	-	-	Wheat		205	-	-	-	Water		927
14	-	-	-	Woods		1265	-	-	-	-	-	-
15	-	-	-	Buildings-Grass-Trees-Drives		386	-	-	-	-	-	-
16	-	-	-	Stone-Steel-Towers		93	-	-	-	-	-	-
	Total		42776	Total		10249	Total		204542	Total		5211

based on CNN architectures such as the pResNet [50], DSNet [51], and S3ANet [52], as well as deep learning methods based on Transformer architectures, including LSFA [53] and MRViT [54]. Additionally, we have included the SSFTT [55] method, which effectively integrates the advantages of both CNNs and Transformers to leverage their complementary strengths. We have also included S2VNet [56], a method that combines nonlinear unmixing with pixel-level classification to enhance classification performance. A concise overview of the comparative methods is provided below, highlighting their key features and contributions.

1) *SVM [7]:* The SVM classifier is implemented using the LIBSVM library, employing the radial basis function kernel.
2) *pResNet [50]:* pResNet leverages transfer learning from a pretrained deep residual network to robustly capture hierarchical features for HSIC.
3) *DSNet [51]:* DSNet integrates advanced deep unmixing techniques specifically designed for HSIC tasks, thereby facilitating the training of deeper networks for improved classification accuracy.
4) *S3ANet [52]:* S3ANet integrates spatial and spectral attention mechanisms to selectively emphasize informative regions and channels in HSIs.
5) *LSFA [53]:* LSFA proposes a local semantic feature aggregation-based Transformer, which enables Transformers to capture long-range dependencies among multiscale features more effectively.
6) *SSFTT [55]:* SSFTT combines a hierarchical CNN module for extracting shallow spectral–spatial features and a Gaussian distribution weighted tokenization module to map these shallow features into high-level semantic representations.
7) *MRViT [54]:* MRViT combines the strengths of Vision Transformers (ViTs) and multiresolution analysis, which adopts a multiscale residual Transformer architecture to learn robust HSI features.
8) *S2VNet [56]:* S2VNet models spectral variability via subpixel unmixing and pixel-level fusion, enhancing classification performance under limited samples. It combines abundance-aware features with spectral correlation for robust HSI feature representation, which enables it to achieve state-of-the-art performance.

For a fair comparison, we randomly repeated each experiment ten times and reported the mean values and standard deviations of the results across these experiments. For fair comparison, all experiments in this article used the same hyperparameter settings from the original publications, as shown in Table IV.

3) Implementation Details: To ensure the consistency and reproducibility of the experiments, all tests were conducted on a laptop equipped with an Intel Core i5-12500H processor, 16 GB of RAM, and an NVIDIA GeForce RTX 3050Ti Laptop GPU with 4 GB of video memory. The software environment consists of a 64-bit Windows 11 operating system, Python 3.9, and the PyTorch 1.12 deep learning framework. The training epochs and the training batch size are set to 500 and 64, respectively.

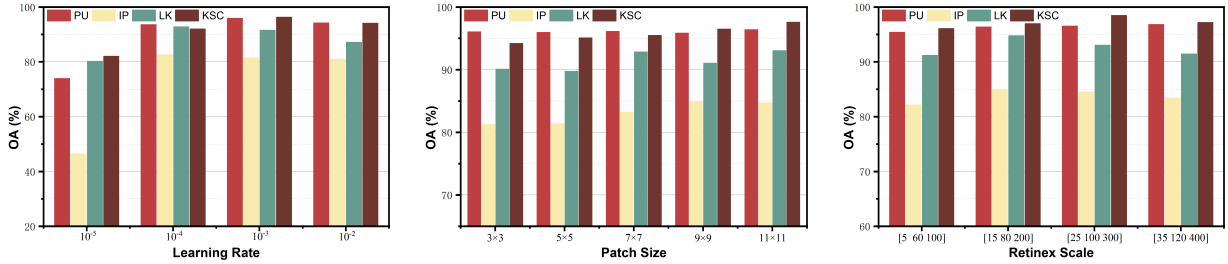
C. Parameter Analysis

In this section, we investigate the settings of three key parameters—lr, patch size, and Retinex scale—and analyze

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TABLE IV
EXPERIMENTAL SETTINGS FOR COMPARISON METHODS

Experiment Settings	pResNet	DSNet	S3ANet	LSFA	SSFTT	MRViT	S2VNet	RDFEN
Training epoch	200	500	1000	100	100	300	500	50
Patch size	7×7	7×7	11×11	15×15	13×13	13×13	7×7	11×11
Batch size	64	64	64	64	64	64	64	64
Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
Learning rate	1×10^{-2}	1×10^{-3}	5×10^{-4}	1×10^{-3}	1×10^{-3}	1×10^{-3}	1×10^{-3}	1×10^{-3}



(a) (b) (c)

Fig. 4. Effect of different parameters on the OA of the model (a) lr, (b) patch size, and (c) Retinex scale.

their impact on the classification performance of our proposed model. As shown in Fig. 4, the influence of each parameter on the OA is demonstrated through individual experiments.

1) Learning Rate: To determine the optimal learning rate (lr) for model training, we tested a range of lrs (lr $\in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$) and evaluated the classification performance across four datasets. As shown in Fig. 4(a), for all four datasets, the classification accuracies of our model tend to first increase and then decrease. Moreover, the PU dataset achieved peak performance at lr = 10^{-3}, while the IP, LK, and KSC datasets yield similar results.

2) Patch Size: The impact of patch size on classification accuracy is evaluated using five different sizes: 3×3, 5×5, 7×7, 9×9, and 11×11. The classification accuracies are presented in Fig. 4. For the PU and LK datasets, the OA reaches its peak with a patch size of 11 ×. For the IP and KSC dataset, the OA similarly achieves satisfying classification performance at the 11×11 patch size.

3) Retinex-Scale: The choice of three scales in the HR module is inspired by the principles of multiscale Retinex algorithms (e.g., MSR [40] and MSRCR [39]). These methods typically use 3–5 scales to balance the capture of local details (small scales) and global structures (large scales). This hierarchical decomposition aligns with human visual perception, where multiple spatial frequencies are processed simultaneously. The fixed number of three scales achieves optimal performance while maintaining computational efficiency, as further increases in scales (e.g., $N_s = 4, 5$) would degrade performance while substantially increasing computational cost.

To validate the choice of $N_s = 3$, we conducted a sensitivity analysis on the PU dataset testing $N_s \in 2, 3, 4, 5$. The results in Table V demonstrate that $N_s = 3$ achieves the highest OA of 96.85%, outperforming other configurations. Reducing N_s to 2 degrades performance substantially, indicating that fewer scales inadequately capture contextual information.

TABLE V
CLASSIFICATION PERFORMANCE AND COMPUTATIONAL COST ACROSS DIFFERENT SCALE NUMBERS (N_s)

N_s	OA (%)	AA (%)	κ (×100)	Training Time (s)
2	95.22±1.18	94.11±1.26	93.70±1.95	632.45
3	**96.85±0.84**	**96.43±1.39**	**95.83±1.12**	714.56
4	96.12±1.02	95.33±1.34	94.71±1.06	812.73
5	95.68±1.83	94.89±0.97	94.33±2.31	968.31

Increasing N_s beyond 3 degrades classification performance while increasing computational cost.

In the RDFEN framework, Retinex processing is applied to HSIs at multiple scales, extracting features from three distinct scales. To evaluate the impact of scale parameter λ, four scale configurations are tested: [5, 60, 100], [15, 80, 200], [25, 100, 300], and [35, 120, 400], with the obtained results presented in Fig. 4(c). For the PU dataset, the model achieves optimal performance with large-scale Retinex settings (λ = [35, 120, 400]). For the IP, LK, and KSC datasets, the highest accuracy is attained under medium-scale configurations (e.g., λ = [25, 100, 300]). This is attributed to the structural complexity of the PU dataset (urban scenes with buildings and roads), necessitating large-scale Retinex processing to capture broad spatial–contextual patterns. In contrast, the IP and LK datasets (agricultural scenes) benefit from medium-scale processing, which better retains fine-grained edge and texture details of crops and vegetation.

D. Ablation Study

To systematically evaluate the performance impact of each model, ablation experiments were conducted in this section, obtaining the results presented in Table VI. The proposed model consists of three key modules: IAFE, RAFE, and IRAIAF, which are analyzed separately.

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TABLE VI

ABLATION EXPERIMENTS CONDUCTED ON THE FOUR DATASETS

IAFE	RAFE	IRAIAF	PU			IP			LK			KSC		
			OA (%)	AA (%)	κ (×100)	OA (%)	AA (%)	κ (×100)	OA (%)	AA (%)	κ (×100)	OA (%)	AA (%)	κ (×100)
✓	×	×	69.00±1.76	66.69±1.29	60.28±1.76	74.37±1.28	82.43±0.70	71.19±1.39	81.70±1.13	84.06±1.31	77.09±1.31	85.24±1.31	83.95±1.74	85.09±0.97
×	✓	×	86.07±1.57	85.32±1.79	81.42±2.11	67.35±2.13	78.12±1.53	63.00±2.39	90.39±1.62	92.11±1.60	89.69±1.86	79.69±3.26	74.18±2.37	77.42±1.94
✓	✓	×	94.63±1.04	94.28±1.25	92.91±1.39	82.74±1.52	87.67±0.97	80.66±1.72	91.78±1.28	95.47±0.63	91.50±1.57	95.31±0.78	95.07±1.21	95.26±0.84
✓	✓	✓	**96.85±0.84**	**96.43±1.39**	**95.83±1.12**	**84.93±1.80**	**89.85±1.01**	**82.88±2.03**	**94.80±0.58**	**97.08±0.43**	**93.28±0.74**	**98.62±0.35**	**97.61±0.45**	**98.46±0.39**

TABLE VII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE PU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Class No.	Training	SVM TGRS2004	pResNet TGRS2019	DSNet TGRS2024	S3ANet TGRS2024	LSFA TGRS2022	SSFTT TGRS2022	MRViT ICCT2022	S2VNet TGRS2025	RDFEN Ours
1	10	67.38±3.72	72.41±3.82	71.06±6.42	66.43±6.36	81.32±2.03	78.48±5.56	69.35±5.79	85.80±3.28	**95.34±2.18**
2	10	62.86±10.7	84.27±2.04	81.08±8.01	89.67±2.35	82.17±3.11	86.55±8.00	84.21±4.52	87.46±3.02	**98.10±1.46**
3	10	51.88±13.1	23.49±4.03	86.95±7.43	64.52±8.42	63.88±4.12	87.32±3.94	**91.26±1.78**	88.63±7.49	90.36±5.54
4	10	88.03±8.66	87.51±5.11	86.38±5.75	69.84±5.66	93.63±2.25	87.80±7.19	86.26±3.89	92.24±4.81	**96.10±2.00**
5	10	99.02±0.40	91.63±6.33	99.81±0.26	83.68±3.65	99.76±0.71	99.85±0.44	99.99±0.02	99.41±0.82	**100.0±0.00**
6	10	68.64±11.1	53.11±11.6	92.53±4.82	83.35±3.47	91.21±2.44	89.78±3.60	93.98±3.28	79.86±9.32	**98.89±1.54**
7	10	90.69±3.14	45.14±6.54	98.94±1.36	93.74±5.12	95.60±2.73	**99.38±0.49**	98.86±0.77	92.29±2.75	97.11±1.60
8	10	78.30±9.00	47.63±10.9	63.58±1.04	91.79±2.79	65.84±4.97	71.79±8.62	51.69±8.40	84.53±9.70	**92.93±3.02**
9	10	**99.91±0.06**	92.46±4.05	91.36±5.84	78.13±4.07	53.15±8.53	96.84±1.50	91.35±3.77	98.40±2.72	99.02±2.02
OA (%)		69.64±3.72	70.80±2.01	81.40±2.99	82.54±1.40	81.95±1.25	85.58±3.81	81.84±2.00	88.03±2.52	**96.85±0.84**
AA (%)		78.52±1.44	66.41±2.85	85.74±1.39	77.02±1.79	80.73±0.79	88.64±1.98	85.22±1.01	90.12±2.43	**96.43±1.39**
κ (×100)		62.13±3.84	60.92±2.81	76.24±3.53	80.13±1.58	76.86±1.47	81.40±4.62	76.56±2.37	84.45±1.53	**95.83±1.12**
Training Time (s)		**0.15**	1479.81	841.27	758.37	8.09	1.77	246.01	614.21	714.56
Test Time (s)		2.95	18.77	6.85	**1.26**	2.86	1.75	185.69	33.65	27.26

1) Ablation of IAFE: IAFE plays an important role in extracting illumination-invariant features from HSIs. As shown in Table VI, ablating the IAFE component causes significant performance degradation in classification accuracy across all benchmark datasets, particularly in urban scenes (with OA decreasing by 27.85% points in the PU dataset). This demonstrates that the IAFE module has distinct advantages in processing urban hyperspectral datasets.

2) Ablation of RAFE: RAFE serves as an inherent attributes feature extractor in the proposed model, aiming to capture the intrinsic property of HSIs. As illustrated in Table VI, without the RAFE module, classification accuracies exhibit significant declines, particularly in the IP and PU datasets (with OA decreasing by 17.68% and 10.78% points, respectively).

3) Ablation for IRAIAF: IRAIAF facilitates the model to learn cross-branch dependencies through adaptive feature fusion. The ablation study in Table VI demonstrates that removing IRAIAF leads to consistent accuracy degradation, validating its necessity in maintaining discriminative feature representations.

In conclusion, the IAFE module, designed to learn illumination-invariant representations, effectively mitigates illumination variability challenges in complex environments (e.g., shadowed regions or low-light scenes). Additionally, the RAFE module further improves classification robustness by extracting discriminative inherent features.

E. Comparative Analysis of Different Methods

Tables VII–X record the quantitative classification results obtained for the PU, IP, LK, and KSC datasets, respectively. When using only ten training samples per class, RDFEN achieved the highest accuracy, significantly outperforming the traditional CNN and Transformer methods, which demonstrates its superiority in scenes with limited training samples. This model can solve the problem of small labeled samples in the following aspects. Illumination variations (e.g., shadows from complex structures) can interfere with the accurate representation of samples, especially when the number of training samples is scarce; this issue becomes even more prominent. The proposed method can effectively decouple illumination variations, which is very helpful to solve the problem of limited samples. Moreover, the dual-branch structure, by integrating intrinsic and extrinsic information, extracts more discriminative features from HSIs, thereby reducing the dependence of classification models on sample size. For the PU dataset, the OA, AA, and κ of our method reach 96.85%, 96.43%, and 95.83%, respectively, significantly outperforming the comparative CNN-based and Transformer-based deep learning methods. This is due to the HR module effectively decoupling pervasive illumination variations in urban environments, coupled with an IRAIAF attention mechanism that enhances robustness under limited training samples through synergistic modeling of illumination-invariant and material-discriminative features. For the IP dataset, most comparative methods achieve an OA lower than 80%. In contrast, the proposed method obtains an OA of 84.93%, surpassing MRViT (the second-best method) by 1.29% points. Moreover, other methods exhibit markedly lower CAs for the classes Soybean-mintill (class no. 11) and Soybean-clean (class no. 12). In contrast, the proposed method achieves a CA of 86.29% for Soybean-mintill and 77.77% for Soybean-clean, demonstrating a marked superiority over comparative methods. Similarly, the classification accuracy of our model on the LK

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TABLE VIII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE IP DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Class No.	Training	SVM TGRS2004	pResNet TGRS2019	DSNet TGRS2024	S3ANet TGRS2024	LSFA TGRS2022	SSFTT TGRS2022	MRViT ICCT2022	S2VNet TGRS2025	RDFEN Ours
1	10	82.22±9.56	25.92±9.56	98.89±2.22	**100.0±0.00**	49.13±14.1	99.44±1.11	**100.0±0.00**	96.94±4.02	**100.0±0.00**
2	10	40.03±5.70	40.33±7.53	44.95±2.57	49.56±2.99	62.54±3.52	62.53±10.4	**89.13±3.35**	62.02±8.90	74.31±3.91
3	10	43.59±8.79	31.15±4.67	25.99±1.32	55.45±3.04	**81.86±4.56**	76.00±6.74	81.46±5.58	74.65±8.23	69.84±5.35
4	10	64.05±8.27	30.96±10.1	90.09±1.97	71.14±2.88	73.67±5.48	96.29±4.00	96.21±1.74	93.13±6.04	**97.14±1.63**
5	10	78.20±9.63	60.72±21.1	51.37±4.69	85.60±2.30	79.33±5.05	**85.98±5.08**	68.98±1.39	85.75±4.26	68.39±0.76
6	10	79.79±6.01	72.35±11.3	81.21±2.12	66.83±2.54	98.27±1.42	96.15±2.70	**99.22±0.55**	96.29±2.70	97.99±0.71
7	10	88.89±4.97	23.23±9.07	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
8	10	77.74±6.84	83.44±12.7	89.72±3.96	98.58±1.87	99.91±0.19	98.18±2.79	**100.0±0.00**	97.56±2.62	**100.0±0.00**
9	10	89.00±17.6	26.00±5.33	90.00±3.00	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
10	10	51.54±10.4	33.56±6.93	49.51±2.19	65.03±9.89	75.74±5.79	**82.56±7.88**	74.86±2.42	78.22±3.50	74.93±4.69
11	10	44.65±7.20	60.94±5.59	55.88±4.36	61.16±4.80	56.92±1.83	68.16±10.15	73.59±2.59	72.25±8.33	**86.29±4.18**
12	10	43.53±6.66	21.21±3.34	38.43±4.27	69.50±3.03	47.04±5.77	63.75±6.83	68.23±6.83	73.02±6.76	**77.77±6.41**
13	10	95.13±3.80	63.80±12.5	93.62±2.68	95.64±1.65	98.53±1.70	99.79±0.61	**100.0±0.00**	98.41±1.75	99.95±0.15
14	10	73.24±8.06	84.73±5.64	62.37±3.95	75.76±0.48	79.43±4.66	91.37±5.73	99.29±0.39	91.91±4.43	**99.81±0.27**
15	10	37.71±8.70	47.80±8.26	85.41±0.85	84.22±4.74	89.01±5.05	**93.37±7.88**	74.68±5.12	85.29±7.84	92.45±5.72
16	10	93.61±2.22	67.02±14.7	97.47±0.32	95.66±1.44	92.79±8.32	98.67±2.60	**100.0±0.00**	99.88±0.36	98.80±0.01
OA (%)		55.49±2.71	50.33±3.76	57.21±4.97	67.30±2.08	72.18±1.19	78.87±3.09	83.64±1.18	78.14±2.23	**84.93±1.80**
AA (%)		67.68±2.22	48.32±2.93	71.96±2.15	63.31±2.27	67.76±0.80	88.26±1.09	89.10±0.67	87.83±2.01	**89.85±1.01**
κ (×100)		50.17±2.93	44.35±3.91	52.67±7.91	79.63±1.31	68.76±1.32	76.22±3.40	81.42±1.32	77.64±2.43	**82.88±2.03**
Training Time (s)		**0.31**	265.80	200.13	799.03	5.41	4.47	397.32	294.09	617.60
Test Time (s)		0.76	19.51	0.94	1.41	1.85	**0.66**	37.95	14.73	17.11

TABLE IX

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE LK DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Class No.	Training	SVM TGRS2004	pResNet TGRS2019	DSNet TGRS2024	S3ANet TGRS2024	LSFA TGRS2022	SSFTT TGRS2022	MRViT ICCT2022	S2VNet TGRS2025	RDFEN Ours
1	10	85.15±6.39	91.49±5.22	98.14±1.22	74.81±7.20	99.19±1.47	98.38±0.83	97.42±1.76	99.28±0.74	**99.35±0.58**
2	10	55.90±10.6	84.96±8.08	94.24±2.92	93.61±3.64	95.63±2.77	95.67±2.63	93.21±3.85	96.21±1.09	**98.27±2.47**
3	10	71.60±10.1	56.73±27.1	98.80±1.51	83.42±5.71	95.55±5.36	**99.51±0.69**	98.83±1.60	95.70±0.75	99.46±0.82
4	10	66.17±7.52	**95.94±2.27**	81.45±9.78	78.38±3.62	90.81±2.75	89.26±3.90	83.53±3.42	90.90±0.98	85.13±1.84
5	10	67.96±10.9	47.30±9.49	87.14±13.5	94.49±2.57	69.12±6.89	97.92±1.31	98.26±1.04	92.91±1.28	**99.90±0.15**
6	10	92.42±4.37	86.08±7.25	93.30±4.86	90.04±4.49	85.85±4.41	96.60±3.86	97.29±2.70	98.79±1.21	**99.24±0.56**
7	10	**99.91±0.05**	98.99±0.57	95.65±1.21	97.48±4.81	97.74±1.38	97.85±0.57	99.09±0.68	99.01±0.24	99.57±0.29
8	10	78.30±10.4	74.35±5.85	72.34±15.1	87.14±2.06	64.91±16.1	86.14±8.38	85.35±2.31	93.28±1.05	**95.20±1.09**
9	10	56.96±6.07	80.59±6.94	88.67±9.30	94.69±4.50	65.54±6.00	92.43±3.50	83.28±5.60	94.19±0.86	**97.56±0.77**
OA (%)		81.84±2.72	91.33±1.71	90.37±2.99	86.46±2.58	92.49±1.20	94.60±1.39	92.75±1.01	93.88±2.71	**94.80±0.58**
AA (%)		74.93±1.74	79.61±3.71	89.97±2.17	82.72±3.18	84.93±2.08	94.86±0.96	92.92±0.82	93.16±0.77	**97.08±0.43**
κ (×100)		77.03±3.23	88.73±2.18	87.66±3.65	88.23±1.99	90.19±1.55	93.00±1.78	90.64±1.26	92.98±2.66	**93.28±0.74**
Training Time (s)		**0.16**	6472.38	1978.55	925.52	40.76	10.04	864.99	1645.31	1038.94
Test Time (s)		15.77	18.88	13.10	**2.05**	2.78	2.58	179.94	161.35	113.87

TABLE X

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE KSC DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Class No.	Training	SVM TGRS2004	pResNet TGRS2019	DSNet TGRS2024	S3ANet TGRS2024	LSFA TGRS2022	SSFTT TGRS2022	MRViT ICCT2022	S2VNet TGRS2025	RDFEN Ours
1	10	91.67±0.78	91.77±1.78	98.24±0.48	97.17±0.86	97.00±1.14	98.11±0.91	96.94±1.65	97.26±0.83	**99.59±0.40**
2	10	93.66±1.97	68.32±1.27	85.92±1.22	84.75±0.53	85.40±0.72	85.02±2.07	92.78±1.35	87.73±3.59	**97.37±2.04**
3	10	87.39±1.30	82.86±1.79	85.71±1.14	87.74±0.70	86.55±1.46	85.21±1.40	87.28±2.81	86.79±2.76	**100.0±0.00**
4	10	57.00±2.63	61.13±2.36	73.71±1.87	64.60±2.17	59.55±2.58	58.30±3.38	65.50±1.38	78.02±3.81	**93.82±2.43**
5	10	59.62±2.27	43.99±4.53	65.71±1.42	55.18±2.89	64.47±2.36	60.55±2.56	69.53±2.64	68.94±7.73	**82.67±1.30**
6	10	67.74±1.93	64.70±2.52	65.29±2.68	50.16±1.98	60.66±1.35	64.61±1.53	69.30±1.23	78.22±5.12	**99.17±1.10**
7	10	72.13±1.14	63.81±1.57	74.67±1.67	72.97±1.72	76.34±1.48	70.58±2.82	69.72±3.77	96.21±2.35	**100.0±0.00**
8	10	57.31±2.09	70.07±1.29	89.43±1.29	79.01±1.45	83.37±1.78	76.75±0.79	80.41±0.22	89.57±1.96	**100.0±0.00**
9	10	76.57±1.34	83.84±2.03	93.64±1.58	89.67±0.74	91.09±0.98	85.07±1.57	91.65±0.99	86.39±2.57	**100.0±0.00**
10	10	71.25±1.74	74.57±1.76	97.05±0.88	97.16±0.67	89.81±1.23	98.40±0.52	93.93±1.96	97.56±1.85	**100.0±0.00**
11	10	71.42±1.28	76.99±0.98	92.94±0.47	85.74±1.61	89.40±0.92	87.93±0.38	80.12±1.28	98.07±2.58	**100.0±0.00**
12	10	96.97±0.58	92.87±1.75	97.49±0.43	95.78±0.81	**97.77±0.72**	83.02±2.38	88.51±1.35	89.07±3.21	96.34±2.57
13	10	98.90±0.35	99.88±0.10	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.87±0.68	99.92±0.23	**100.0±0.00**
OA (%)		79.64±2.09	80.59±1.59	91.61±1.30	86.61±1.66	88.51±1.27	84.69±2.24	87.90±1.12	89.86±1.21	**98.62±0.35**
AA (%)		77.05±1.74	77.36±2.11	86.14±1.71	80.85±2.04	86.35±1.42	78.51±1.47	83.51±1.98	88.67±1.65	**97.61±0.45**
κ (×100)		77.32±1.93	80.06±0.94	90.62±1.09	85.10±1.32	88.25±0.93	82.98±2.34	86.52±1.37	88.69±1.35	**98.46±0.39**
Training Time (s)		**0.12**	242.18	129.53	524.69	7.75	12.96	206.33	319.28	230.24
Test Time (s)		**1.05**	16.33	1.12	1.08	1.19	1.43	28.16	46.65	12.08

dataset is higher than that of other state-of-the-art methods. More notably, the proposed method obtains the highest CA in 6 classes and the second-highest CA in 2 classes (9 classes in total). For the KSC dataset, our method achieves state-of-the-art performance, obtaining the highest accuracy in 12 classes (13 classes in total).

The computational costs of different methods are also investigated. As illustrated in Tables VII–X, the traditional SVM

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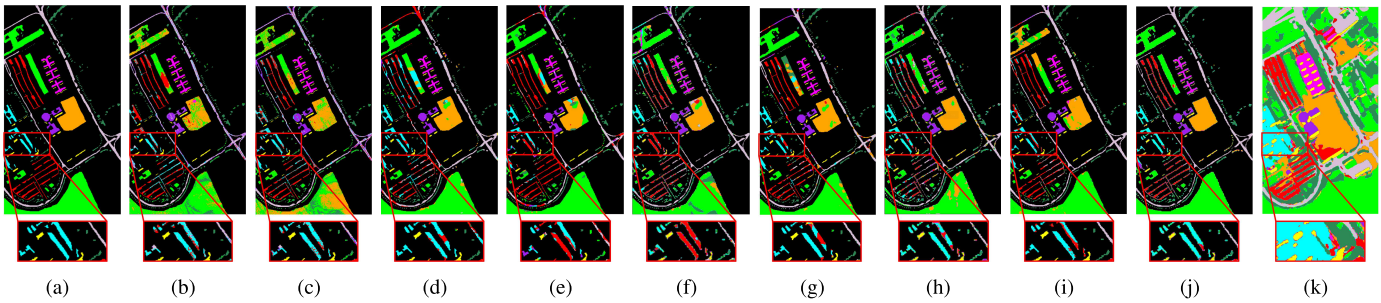


Fig. 5. Classification maps obtained by different methods on the PU dataset. (a) Ground truth. (b) SVM. (c) pResNet. (d) DSNet. (e) S3ANet. (f) LSFA. (g) SSFTT. (h) MRViT. (i) S2VNet. (j) RDFEN (ours). (k) RDFEN (including the unlabeled areas).

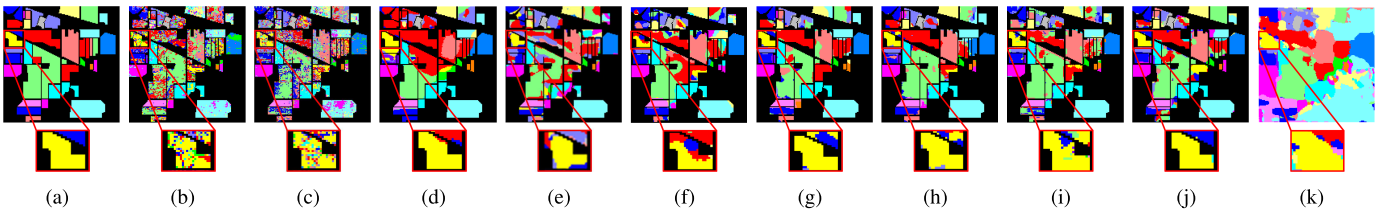


Fig. 6. Classification maps obtained by different methods on the IP dataset. (a) Ground truth. (b) SVM. (c) pResNet. (d) DSNet. (e) S3ANet. (f) LSFA. (g) SSFTT. (h) MRViT. (i) S2VNet. (j) RDFEN (ours). (k) RDFEN (including the unlabeled areas).

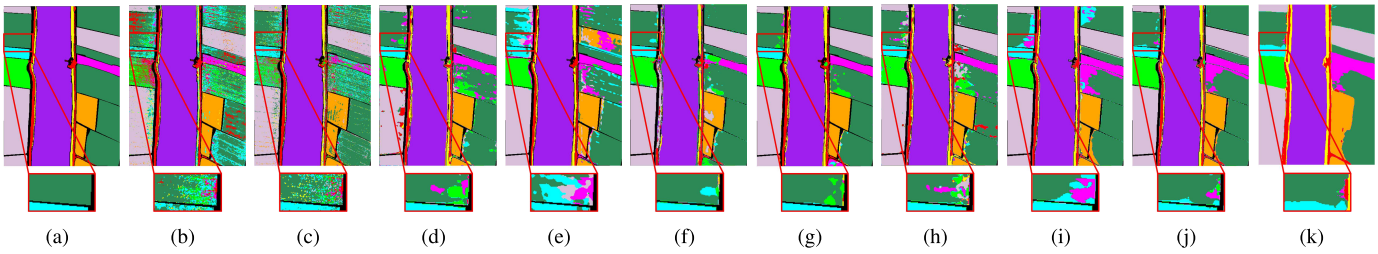


Fig. 7. Classification maps obtained by different methods on the LK dataset. (a) Ground truth. (b) SVM. (c) pResNet. (d) DSNet. (e) S3ANet. (f) LSFA. (g) SSFTT. (h) MRViT. (i) S2VNet. (j) RDFEN (ours). (k) RDFEN (including the unlabeled areas).

method demonstrates better performance in terms of execution speed. Transformer-based methods generally exhibit higher computational efficiency compared to CNN-based methods in processing HSIs. This is because CNNs necessitate numerous layers of convolution operations to progressively reduce spatial dimensions, thereby enabling a feasible computational load. In contrast, Transformers can directly operate on flattened feature vectors via linear transformations, thus avoiding the additional computational overhead associated with dimensionality reduction. Although the computational efficiency of our method is not optimal, it still exhibits greater competitiveness compared to CNN-based methods, such as pResNet, DSNet, and S3ANet.

Figs. 5–8 visualize some of the classification maps generated by all studied methods for the PU, IP, LK, and KSC datasets, respectively. As illustrated in Fig. 5, the SVM struggles to build robust feature mapping mechanisms due to its limited model capacity, and its high sensitivity to data noise and outliers leads to a significant decline in classification performance. Similarly, the CNN-based pResNet model also suffers from sensitivity to noise (especially in the presence of limited samples). Although DSNet and S3ANet exhibit enhanced noise robustness via regularization,

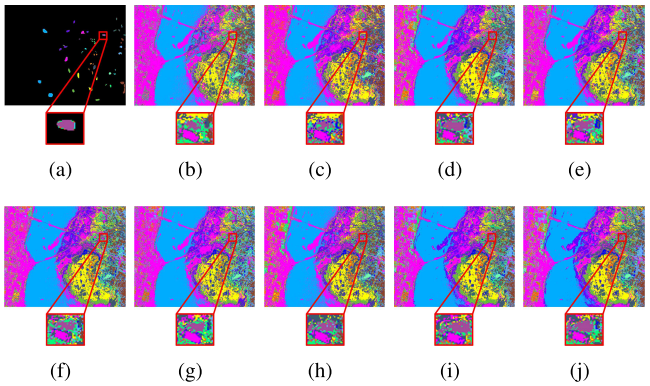
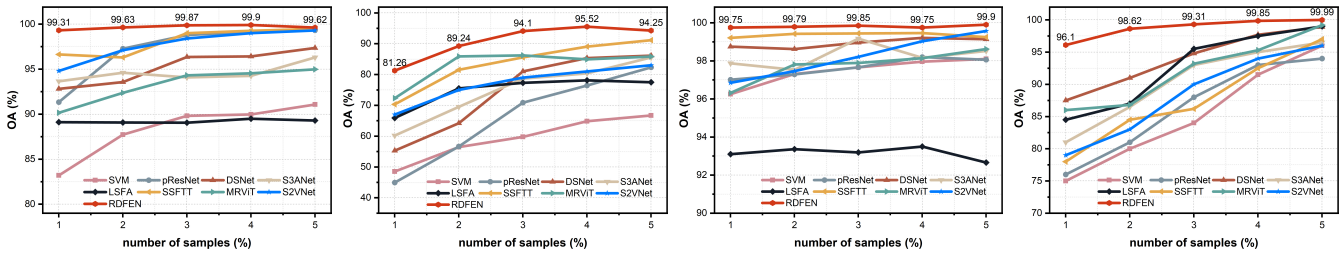


Fig. 8. Classification maps obtained by different methods on the KSC dataset. (a) Ground truth. (b) SVM. (c) pResNet. (d) DSNet. (e) S3ANet. (f) LSFA. (g) SSFTT. (h) MRViT. (i) S2VNet. (j) RDFEN (ours).

overfitting remains an issue in highly similar spectral classes (e.g., meadows and bare soil) due to limited training samples. This can result in misclassifications at boundaries between different classes. While the Transformer architecture effectively models long-range dependencies through self-attention mechanisms, its feature extraction process is quite sensitive

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(a) (b) (c) (d)

Fig. 9. OAs with varying sizes of the training set across the considered datasets. (a) PU. (b) IP. (c) LK. (d) KSC.

TABLE XI
COMPARISON OF DIFFERENT ATTENTION MECHANISMS ACROSS FOUR DATASETS

Attention Mechanism	Pavia University			Indian Pines			WHU-Hi-LongKou			KSC		
	OA (%)	AA (%)	κ ($\times$100)	OA (%)	AA (%)	κ ($\times$100)	OA (%)	AA (%)	κ ($\times$100)	OA (%)	AA (%)	κ ($\times$100)
Interactive Attention	**96.85±0.84**	**96.43±1.39**	**95.83±1.12**	**84.93±1.80**	**89.85±1.01**	**82.88±2.03**	**94.80±0.58**	**97.08±0.43**	**93.28±0.74**	**98.62±0.35**	**97.61±0.45**	**98.46±0.39**
Channel Attention	94.83±2.17	93.41±1.67	92.26±1.64	82.93±1.22	87.44±1.31	80.24±1.57	92.09±0.95	95.34±1.45	91.63±0.92	95.94±1.14	95.47±1.32	95.76±1.03
Spatial Attention	95.31±1.12	95.07±1.19	94.91±1.27	83.57±0.96	88.61±1.34	81.98±1.49	92.91±1.34	96.01±1.13	92.24±1.04	97.21±0.83	96.82±0.91	96.92±1.15

to shifts in the input data distribution. Under drastic illumination changes, slight perturbations in the input data can lead to asymmetric distortions in the Transformer's latent space features, thereby weakening the model's domain adaptation capability. Moreover, influenced by illumination gradients and shadow effects, spectral mixing phenomena at heterogeneous region boundaries exacerbate decision boundary uncertainties for LSFA and MRViT. In contrast, SSFTT leverages a cascaded CNN–Transformer hybrid architecture to achieve collaborative modeling of local details and global semantics, demonstrating significant advantages in classification tasks with complex boundaries. These results indicate that methods based on Transformers often outperform those based on CNNs in terms of interclass separability metrics. More importantly, the proposed RDFEN method (within the CNN framework) achieves superior performance in terms of object integrity and boundary delineation.

F. Robustness Evaluation

To comprehensively evaluate the classification performance and robustness of our model, additional experiments are performed using varying numbers of training samples. For all four datasets, the training sets were constructed by randomly selecting 1%–5% of the total samples. The OAs of all investigated methods across the four datasets are recorded in Fig. 9. As illustrated, with an increase in the size of training data, the classification performance of all examined methods shows consistent improvements. Moreover, our method achieves higher OAs compared to all other methods, no matter the size of the training set. Crucially, the advantages of our method are particularly evident when the number of samples is very limited. This robust performance demonstrates the effectiveness and adaptability of the proposed method in situations with very few samples, exhibiting great potential in practical applications where labeled data are normally very scarce.

G. Comparison With Standard Attention Mechanisms

To validate the effectiveness of the proposed IRAIAF module, we conducted comparative experiments with common channel and spatial attention mechanisms on four benchmark datasets.

As observed in Table XI, IRAIAF consistently outperforms both baselines across all datasets and metrics. For example, on KSC, IRAIAF achieves an OA of 98.62%—significantly higher than channel attention (95.94%) and spatial attention (97.21%). Similar results can be observed for AA and Kappa, with significant improvements across all datasets. These results highlight the robustness of IRAIAF in handling complex illumination variations, particularly in urban scenes where shadows and lighting gradients degrade classification accuracy. The superior performance is attributed to IRAIAF's explicit modeling of cross-branch interactions between illumination and reflectance attributes, enabling dynamic feature fusion and enhanced discriminative power under challenging lighting conditions.

H. Overfitting Analysis of RDFEN

In this section, we perform experiments on the PU dataset by systematically deactivating key modules (IAFE, RAFE, and IRAIAF) and analyzing training/validation loss curves to demonstrate how the full model mitigates overfitting.

As shown in Fig. 10, the full RDFEN achieves a training loss reduction from 2.0 to 0.1 within 50 epochs, with validation loss decreasing from 1.5 to 0.5. In contrast, removing IAFE accelerates training loss reduction but increases validation loss to 1.0, suggesting overfitting is potentially linked to inadequate illumination-invariant feature extraction. Removing RAFE slows convergence, highlighting its role in controlling model complexity through reflectance-focused representation. Removing IRAIAF leads to severe overfitting (validation loss 1.2), where absent cross-branch attention fusion correlates with degraded feature integration. These results indicate that IAFE and IRAIAF implicitly regularize the model by decoupling illumination variations and enabling cross-branch feature fusion, while RAFE stabilizes training by prioritizing discriminative spectral–spatial patterns.

Furthermore, we employ standard Dropout layers for complementary regularization by randomly deactivating neurons

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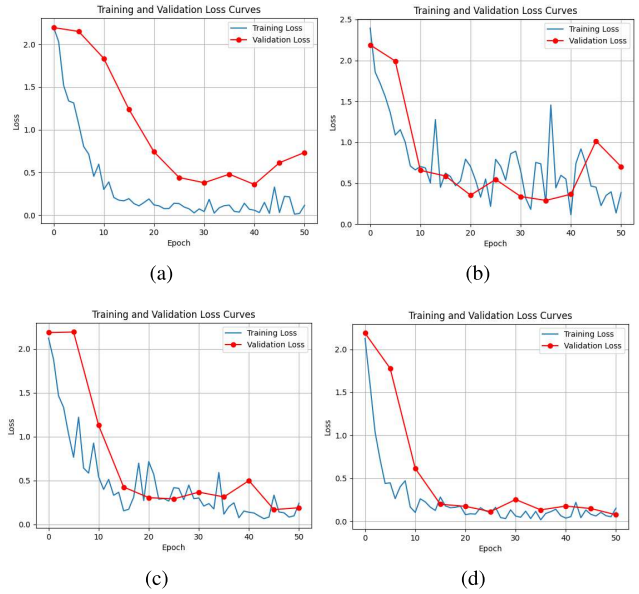


Fig. 10. Train/validation loss curves for ablated configurations. (a) IAFE deactivated. (b) RAFE deactivated. (c) IRAIAF deactivated. (d) Full RDFEN.

TABLE XII
COMPARISON OF PARAMETER COUNT AND FLOPS ON FOUR DATASETS

Datasets	Metric	Models							
		pResNet	DSNet	S3ANet	LSFA	SSFTT	MRViT	S2VNet	RDFEN
PU	Params (M)	1.092	0.342	50.324	0.307	0.154	27.061	0.326	0.272
	FLOPs (G)	0.929	1.661	73.280	0.282	0.016	6.846	0.069	0.021
IP	Params (M)	1.120	0.381	50.082	0.295	0.154	28.073	0.307	0.281
	FLOPs (G)	0.921	1.528	75.442	0.282	0.016	7.159	0.069	0.022
LK	Params (M)	1.123	0.34	50.499	0.335	0.154	28.078	0.307	0.274
	FLOPs (G)	0.928	0.811	75.34	0.282	0.016	7.172	0.071	0.021
KSC	Params (M)	1.127	0.415	50.171	0.293	0.154	28.07	0.307	0.262
	FLOPs (G)	0.925	1.094	75.44	0.282	0.016	7.103	0.068	0.021

during training. These results align with our design philosophy of decoupling illumination and reflectance factors, which improves classification accuracy while enabling more generalizable representations that indirectly mitigate overfitting.

I. Analysis of Computational Complexity

Table XII presents the performance metrics along with computational costs [parameter count and floating point operations (FLOPs)] obtained from various methods across all datasets. The performance of RDFEN is superior to that of most listed methods across the four datasets. While Transformer-based models (MRViT and LSFA) achieve high accuracy, their computational cost often makes them impractical for edge devices. In contrast, RDFEN maintains competitive accuracy while drastically reducing resource consumption. For instance, on the PU dataset, RDFEN achieves 96.85% OA with 0.272M parameters and 0.021G FLOPs, compared to MRViT's 81.84% OA with 27.061M parameters and 6.846G FLOPs. SSFTT has minimal FLOPs due to a shallow architecture, but lower accuracy. In conclusion, RDFEN provides the best accuracy–efficiency balance.

J. Discriminative Capability Assessment

To further evaluate the discriminative ability of our method under varying illumination conditions, we added ROC curves

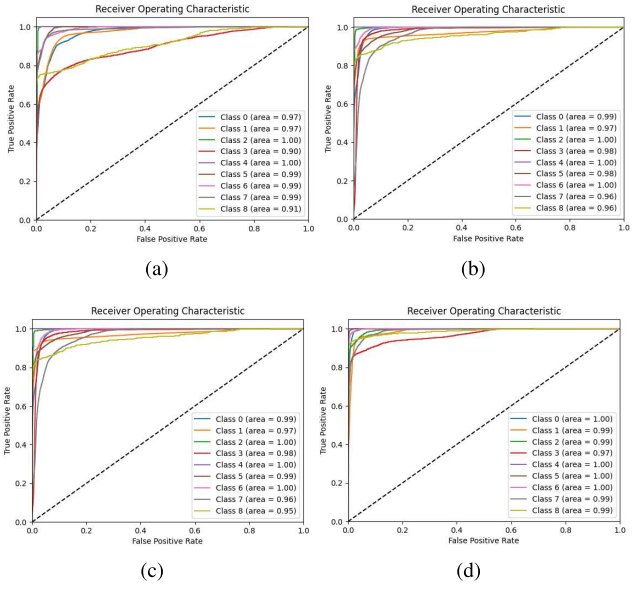


Fig. 11. ROC curves on the PU dataset. (a) LSFA. (b) SSFTT. (c) MRViT. (d) RDFEN.

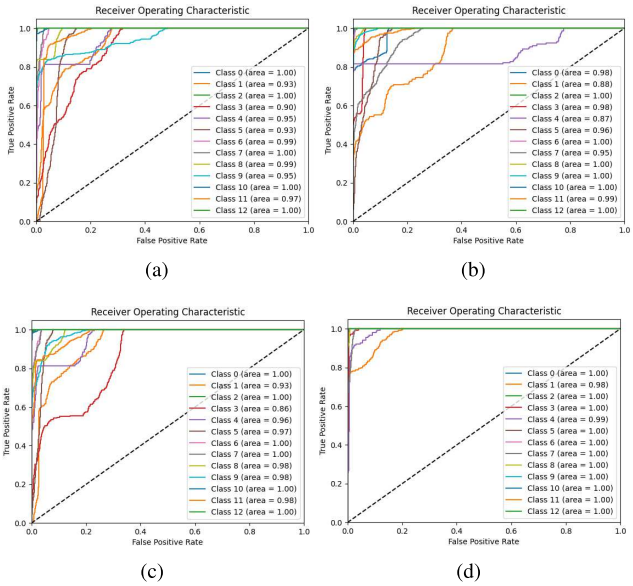


Fig. 12. ROC curves on the KSC dataset. (a) LSFA. (b) SSFTT. (c) MRViT. (d) RDFEN.

for all classes on the PU and KSC datasets. These curves provide a visual representation of the tradeoff between true-positive rates (TPRs) and false-positive rates (FPRs) across different classification thresholds. We selected three high-accuracy methods (LSFA, SSFTT, and MRViT) for comparison. For the PU dataset, as shown in Fig. 11, RDFEN achieved an average AUC of 0.982, significantly outperforming LSFA (0.945), SSFTT (0.938), and MRViT (0.927). This highlights the dual-branch architecture's ability to decouple illumination and reflection features, particularly in shadowed regions (e.g., AUC = 0.975 for the class " Shadows"). For the KSC dataset, as shown in Fig. 12, RDFEN also demonstrated superior performance with an average AUC of 0.986, outperforming MRViT by 2.4% points. These results, consistent with

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OA/AA metrics, validate RDFEN's robustness to complex lighting conditions.

V. Conclusion

In this study, we introduce a new RDFEN for HSIC. The proposed RDFEN model emulates computational principles from the human visual system to compensate for illumination variation in HSIs. Experiments conducted on four benchmark HSIs demonstrate the superiority of our RDFEN over several state-of-the-art methods. Moreover, our method achieves superior performance in urban scenes compared to agricultural regions, validating that RDFEN effectively mitigates illumination variation effects in HSIs. Furthermore, leveraging the dual-branch framework, RDFEN simultaneously extracts the illumination and inherent information, achieving robust classification performance with scarce training data. While primarily designed for classification, the illumination-invariant features extracted by our method could significantly advance hyperspectral interpretation tasks such as object detection or semantic segmentation under varying illumination. Future work will also explore: 1) adaptive decomposition mechanisms to handle complex illumination gradients in large-scale HSIs and 2) integration of learnable spectral attention within the Retinex decomposition for discriminative band enhancement.

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